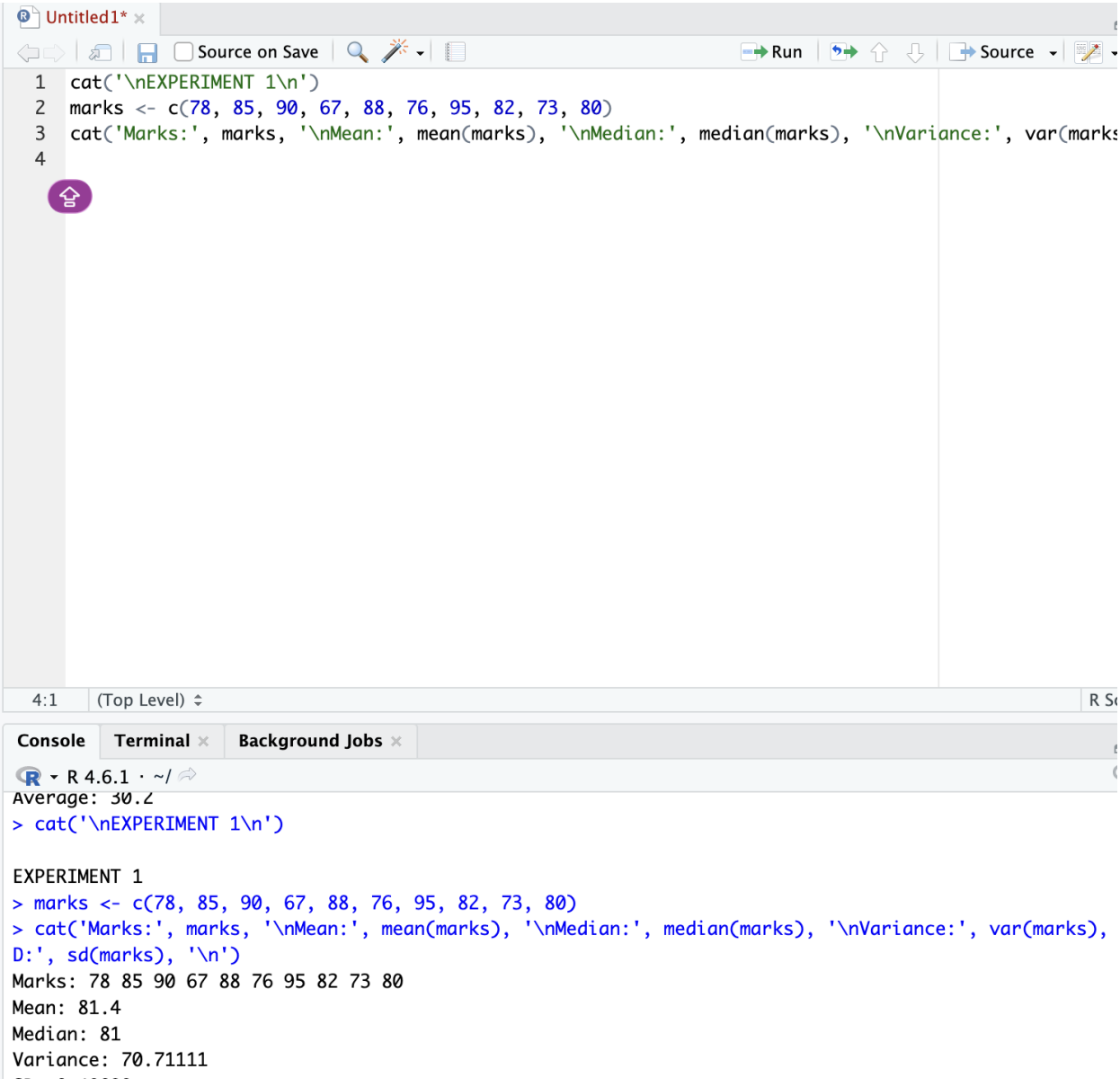
## EXPERIMENT 1: Descriptive Statistics of Student Marks

### AIM:

To calculate the mean, median, variance and standard deviation of a set of marks using R.

### OUTPUT:




### RESULT:

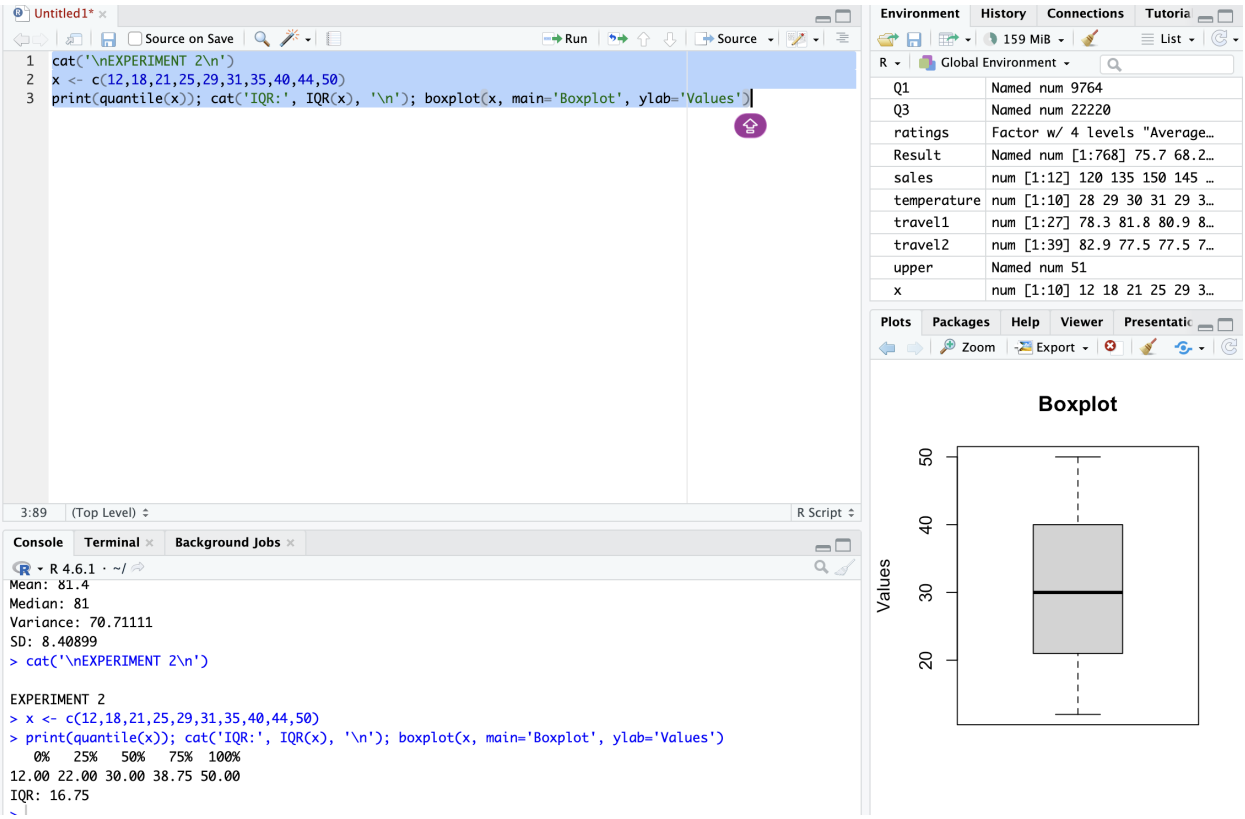
The mean, median, variance and standard deviation of the marks were successfully calculated using R.

## EXPERIMENT 2: Quartiles, Interquartile Range and Boxplot

### AIM:

To find the quartiles and interquartile range (IQR) of a data set and draw a boxplot using R.

### OUTPUT:




### RESULT:

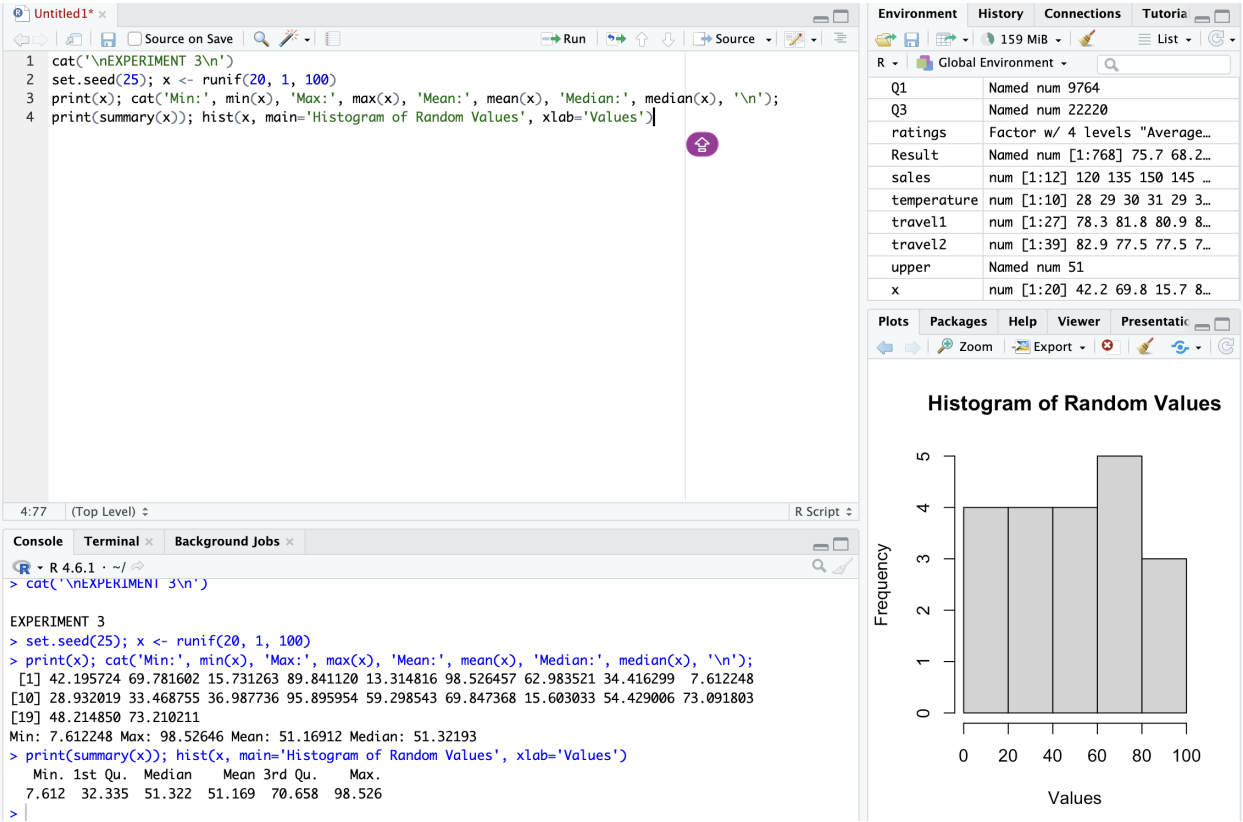
The quartiles and IQR were successfully calculated and the boxplot was successfully drawn using R.

## EXPERIMENT 3: Summary Statistics and Histogram of Random Values

### AIM:

To generate random values, find their minimum, maximum, mean, median and summary, and draw a histogram using R.

### OUTPUT:




### RESULT:

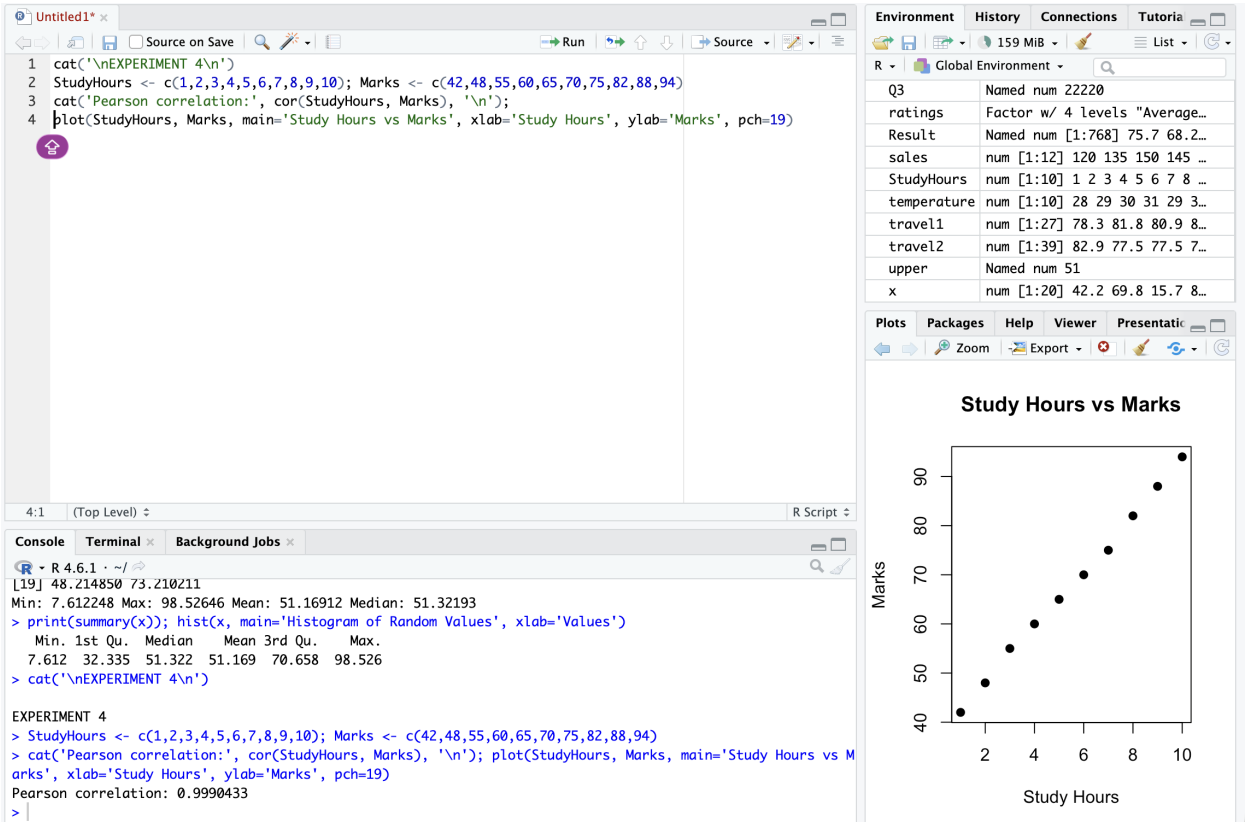
The summary statistics were successfully computed and the histogram was successfully plotted using R.

## EXPERIMENT 4: Correlation and Scatter Plot

### AIM:

To find the Pearson correlation between study hours and marks and draw a scatter plot using R.

### OUTPUT:




### RESULT:

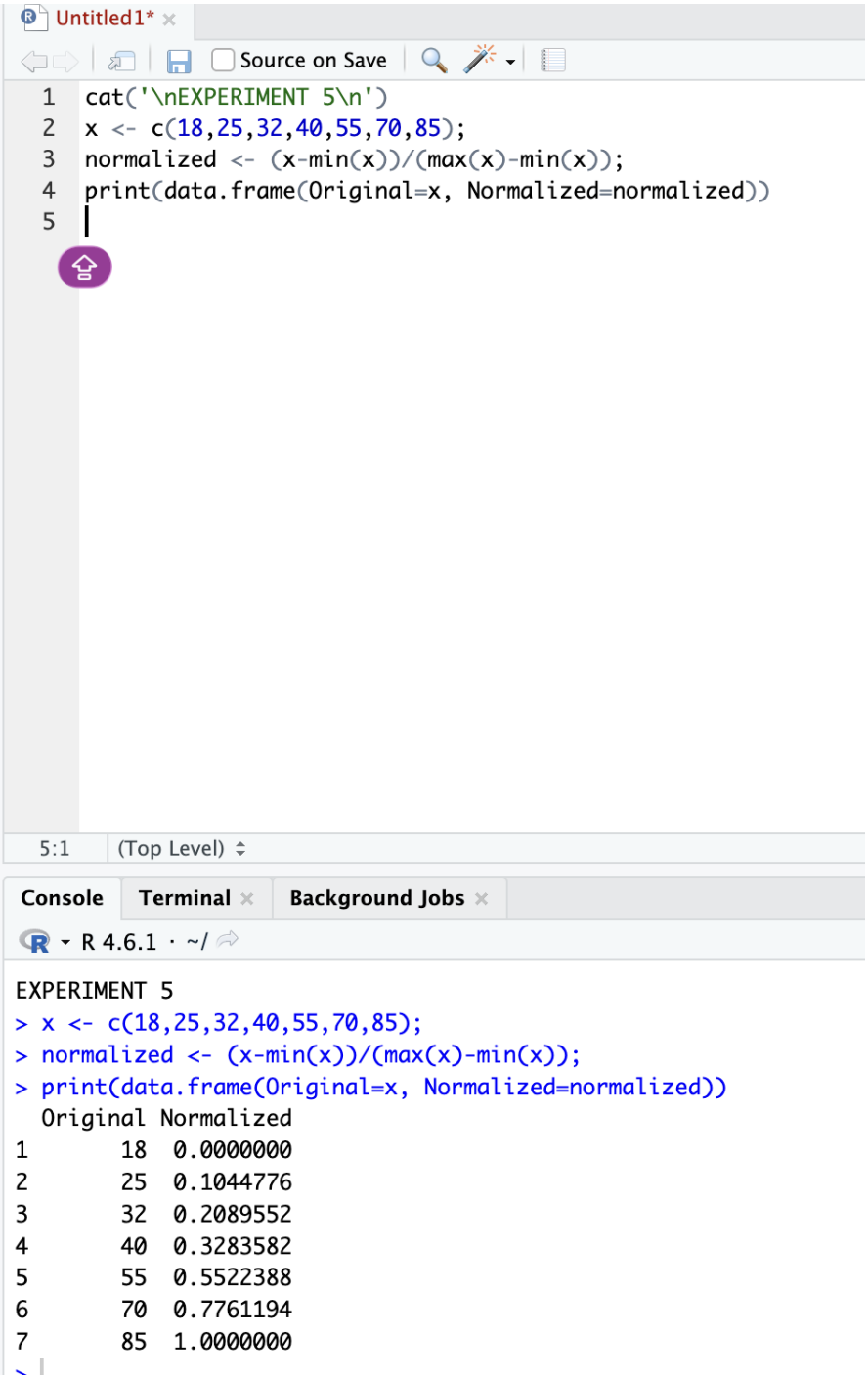
The Pearson correlation was successfully calculated and the scatter plot was successfully drawn using R.

## EXPERIMENT 5: Min-Max Normalization

### AIM:

To normalize a set of values to the range 0 to 1 using min-max normalization in R.

### OUTPUT:




### RESULT:

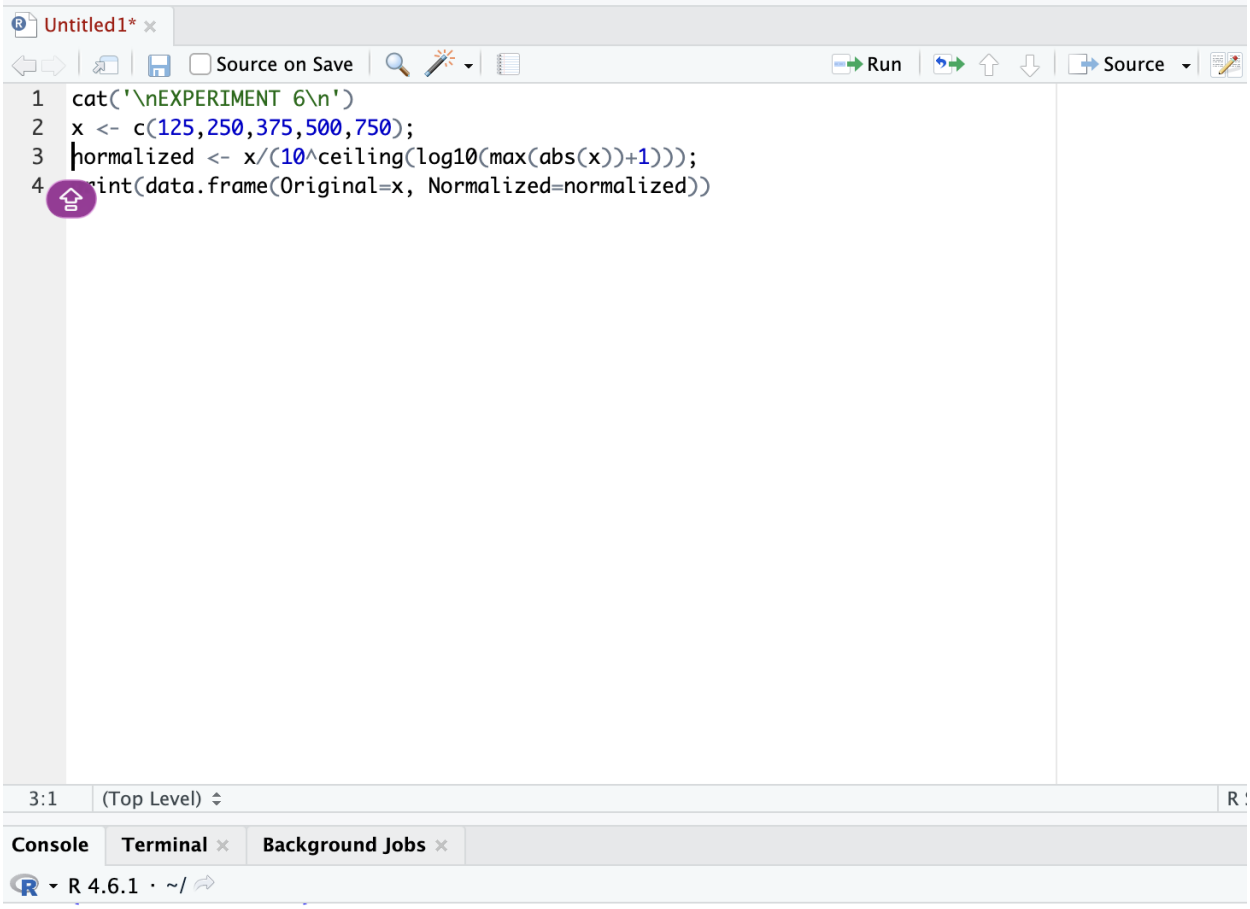
Min-max normalization was successfully performed on the data using R.

## EXPERIMENT 6: Decimal Scaling Normalization

### AIM:

To normalize a set of values using decimal scaling normalization in R.

### OUTPUT:




```
EXPERIMENT 6
> x <- c(125,250,375,500,750); normalized <- x/(10^ceiling(log10(max(abs(x))+1))); print(data.frame(O
l=x, Normalized=normalized))
  Original Normalized
1      125      0.125
2      250      0.250
3      375      0.375
4      500      0.500
5      750      0.750
> cat('\nEXPERIMENT 7\n')
```

### RESULT:

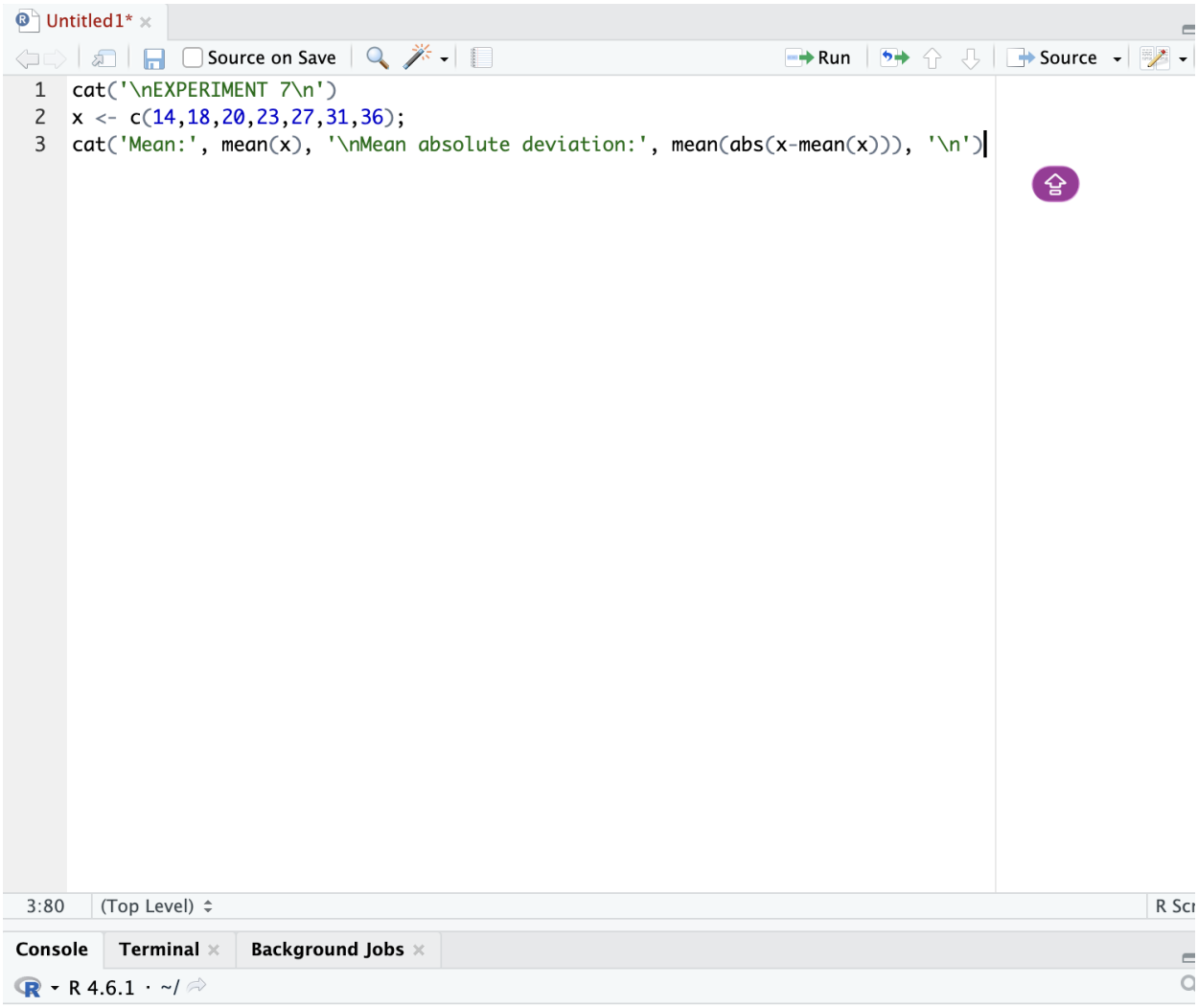
Decimal scaling normalization was successfully performed on the data using R.

## EXPERIMENT 7: Mean and Mean Absolute Deviation

### AIM:

To calculate the mean and the mean absolute deviation of a data set using R.

### OUTPUT:




### RESULT:

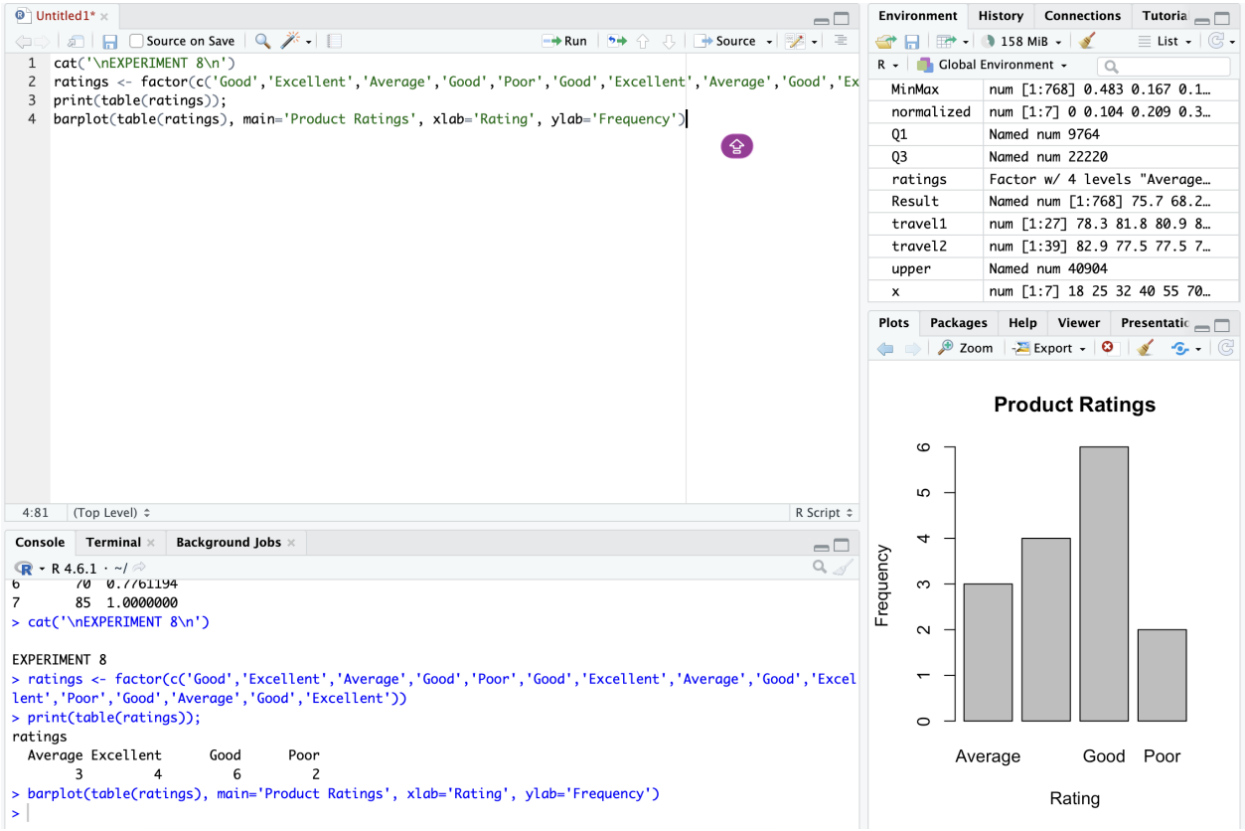
The mean and the mean absolute deviation were successfully calculated using R.

## EXPERIMENT 8: Frequency Distribution and Bar Plot for Categorical Data

### AIM:

To find the frequency distribution of categorical data and display it as a bar plot using R.

### OUTPUT:




### RESULT:

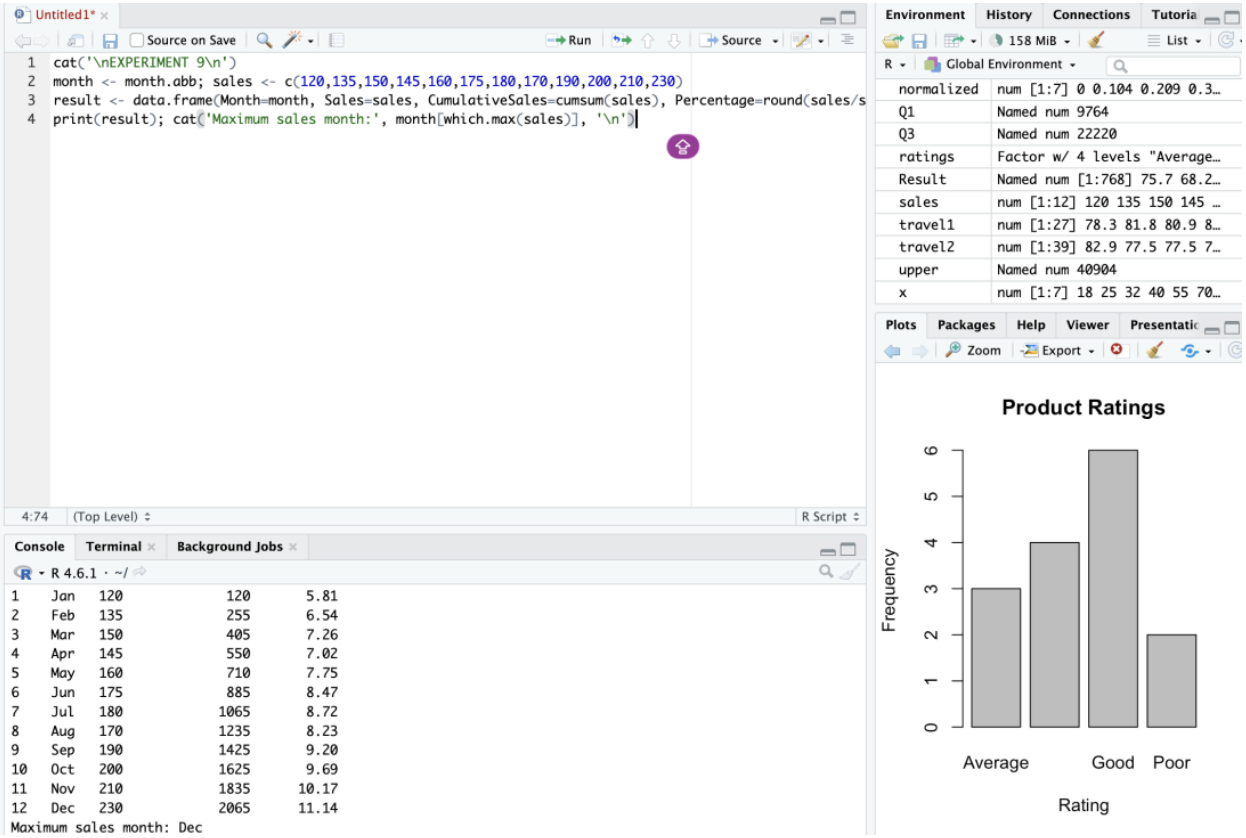
The frequency distribution was successfully obtained and the bar plot was successfully drawn using R.

## EXPERIMENT 9: Monthly Sales Analysis Using Cumulative Sales and Percentage Contribution

### AIM:

To analyse monthly sales by finding the cumulative sales and the percentage contribution of each month using R.

### OUTPUT:




### RESULT:

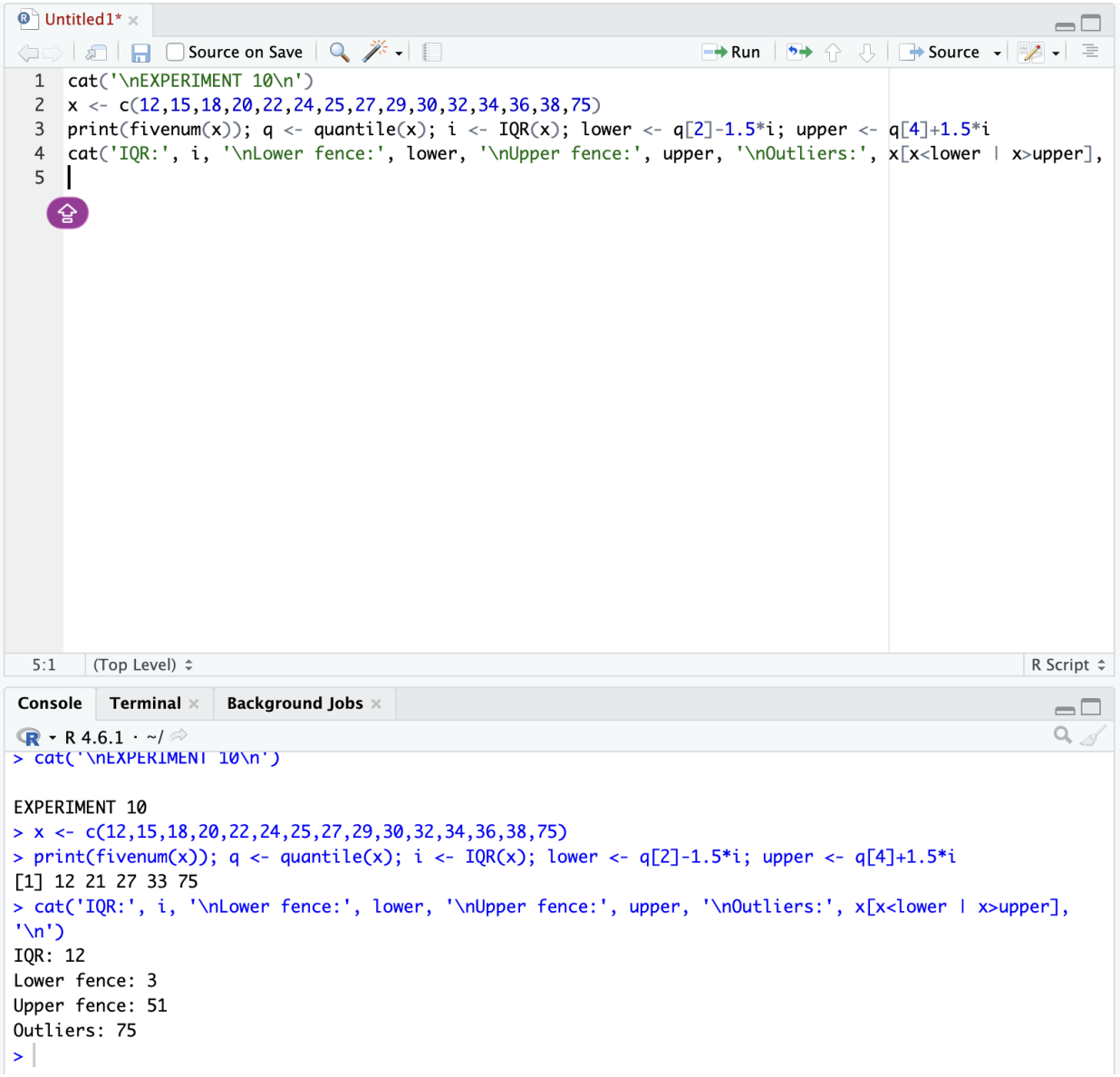
The cumulative sales, percentage contribution and the month with maximum sales were successfully obtained using R.

## EXPERIMENT 10: Five-Number Summary and Outlier Detection Using IQR

### AIM:

To find the five-number summary of a data set and detect outliers using the IQR method in R.

### OUTPUT:




### RESULT:

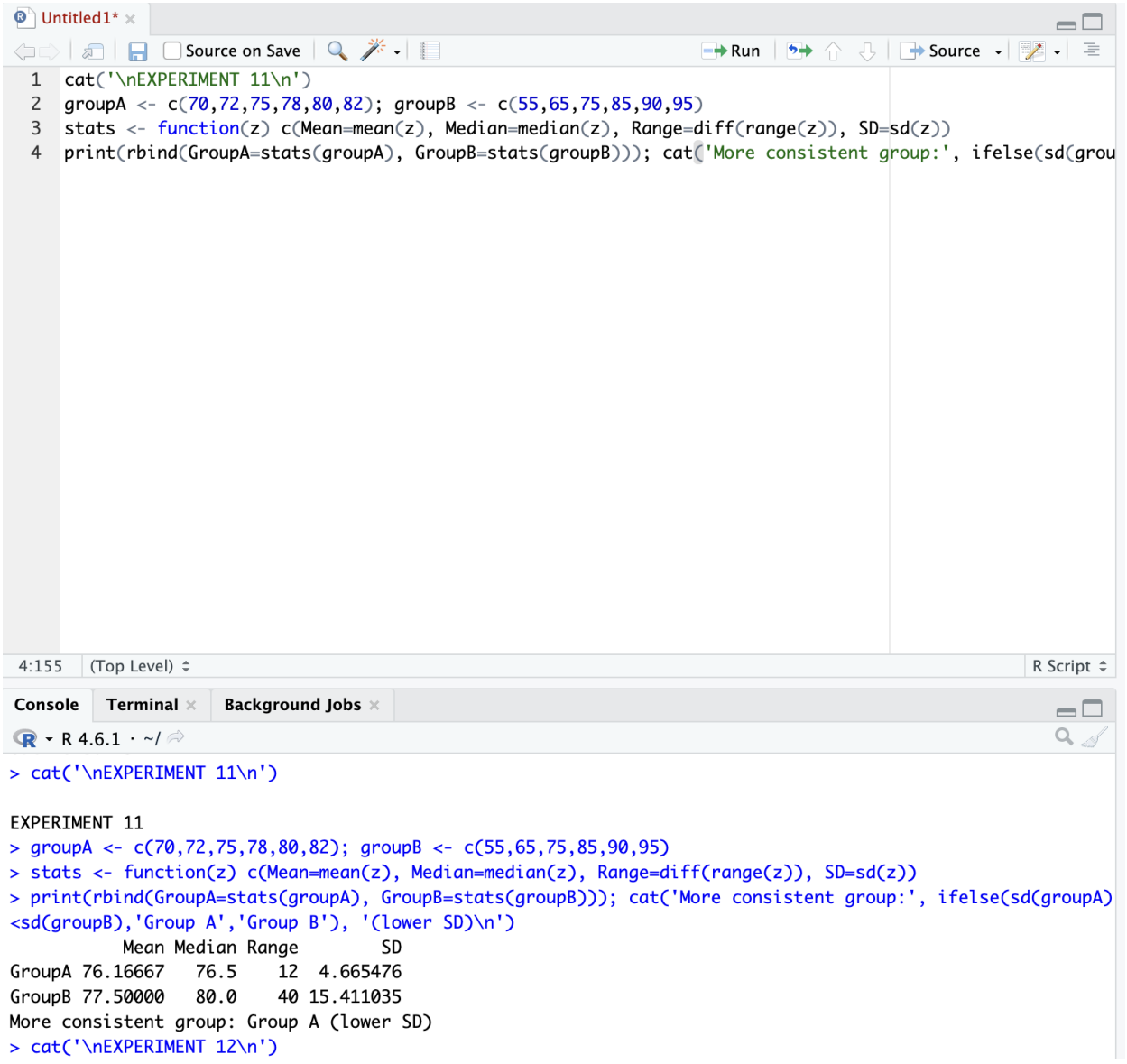
The five-number summary and the fences were successfully calculated and the outlier was successfully detected using R.

## EXPERIMENT 11: Comparison of Two Groups Using Mean, Median, Range and Standard Deviation

### AIM:

To compare two groups of data using mean, median, range and standard deviation and find the more consistent group using R.

### OUTPUT:




### RESULT:

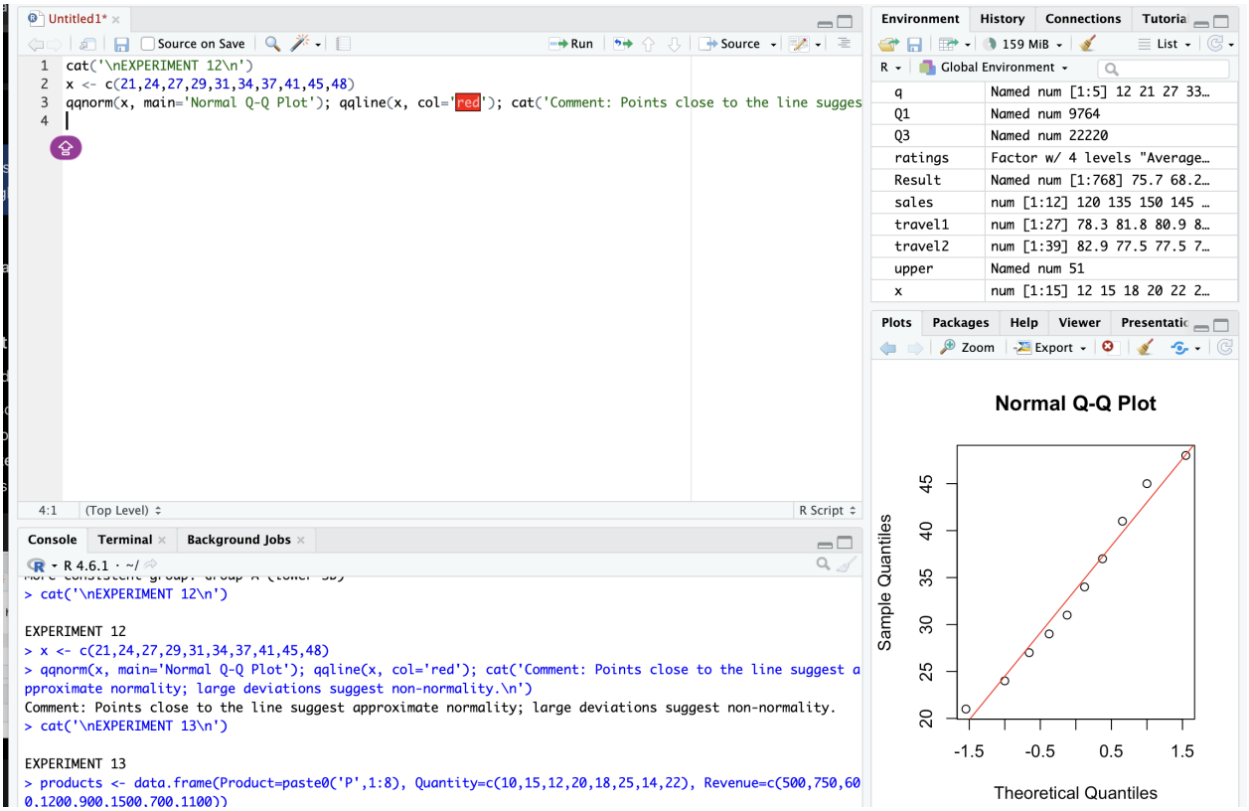
The descriptive statistics of both groups were successfully calculated and compared using R.

## EXPERIMENT 12: Normal Q-Q Plot to Check Normality

### AIM:

To draw a normal Q-Q plot and check whether the data is approximately normally distributed using R.

### OUTPUT:




### RESULT:

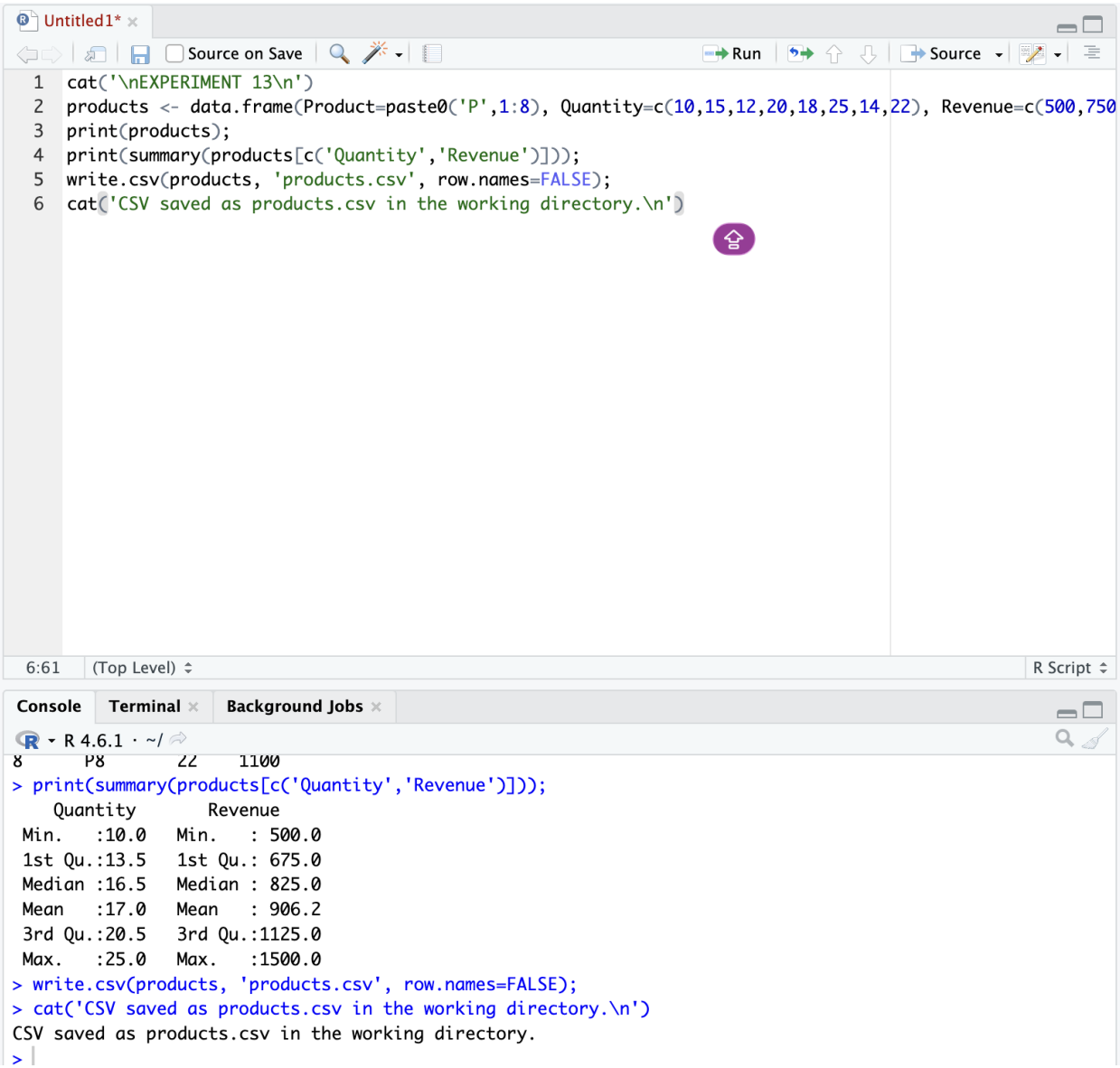
The normal Q-Q plot was successfully drawn using R and the normality of the data was checked.

## EXPERIMENT 13: Data Frame Creation, Summary and Export to CSV

### AIM:

To create a data frame of products, find the summary of quantity and revenue, and save the data as a CSV file using R.

### OUTPUT:




### RESULT:

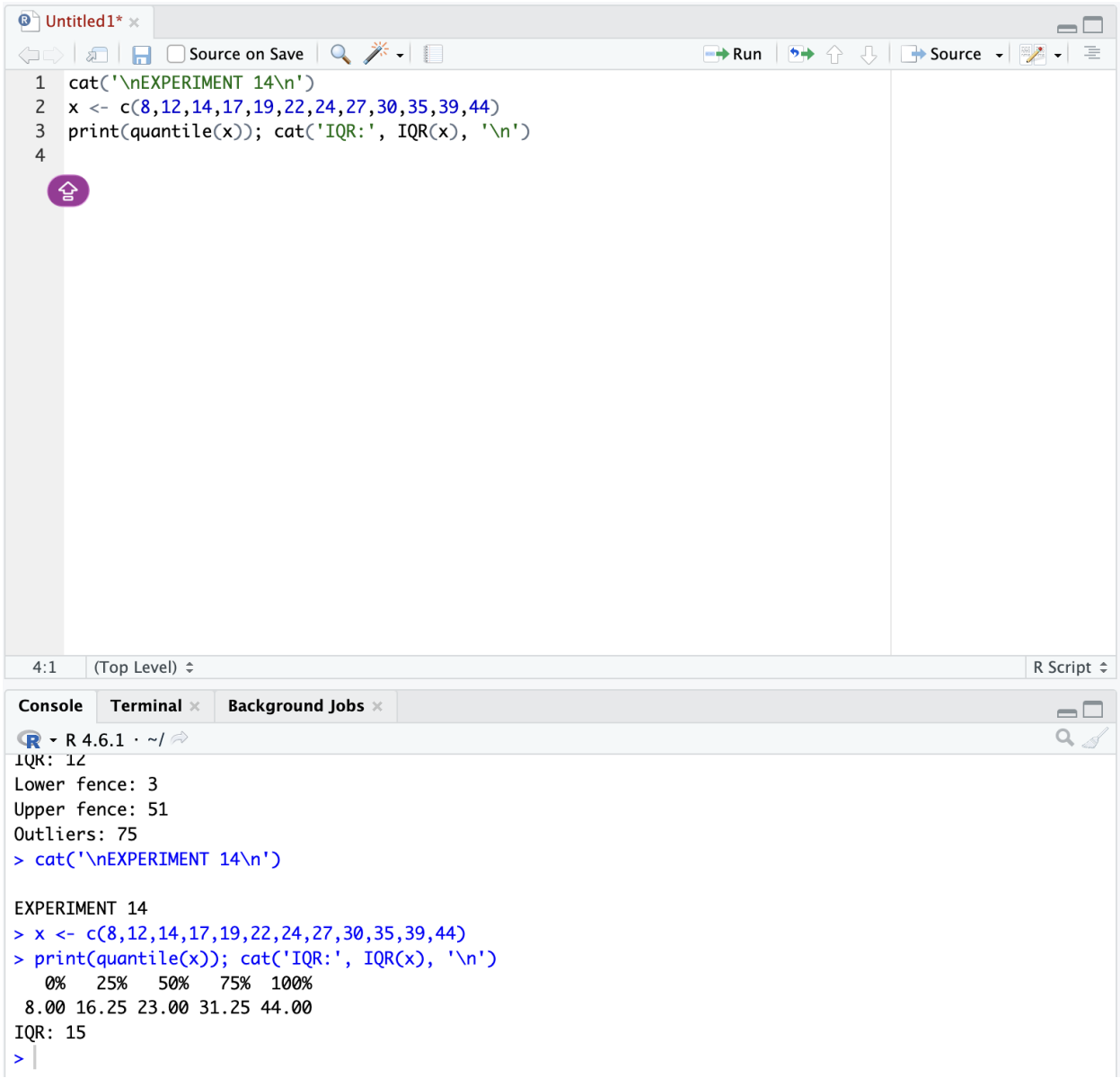
The data frame was successfully created, summarized and saved as a CSV file using R.

## EXPERIMENT 14: Quartiles and Interquartile Range

### AIM:

To calculate the quartiles and the interquartile range of a data set using R.

### OUTPUT:




### RESULT:

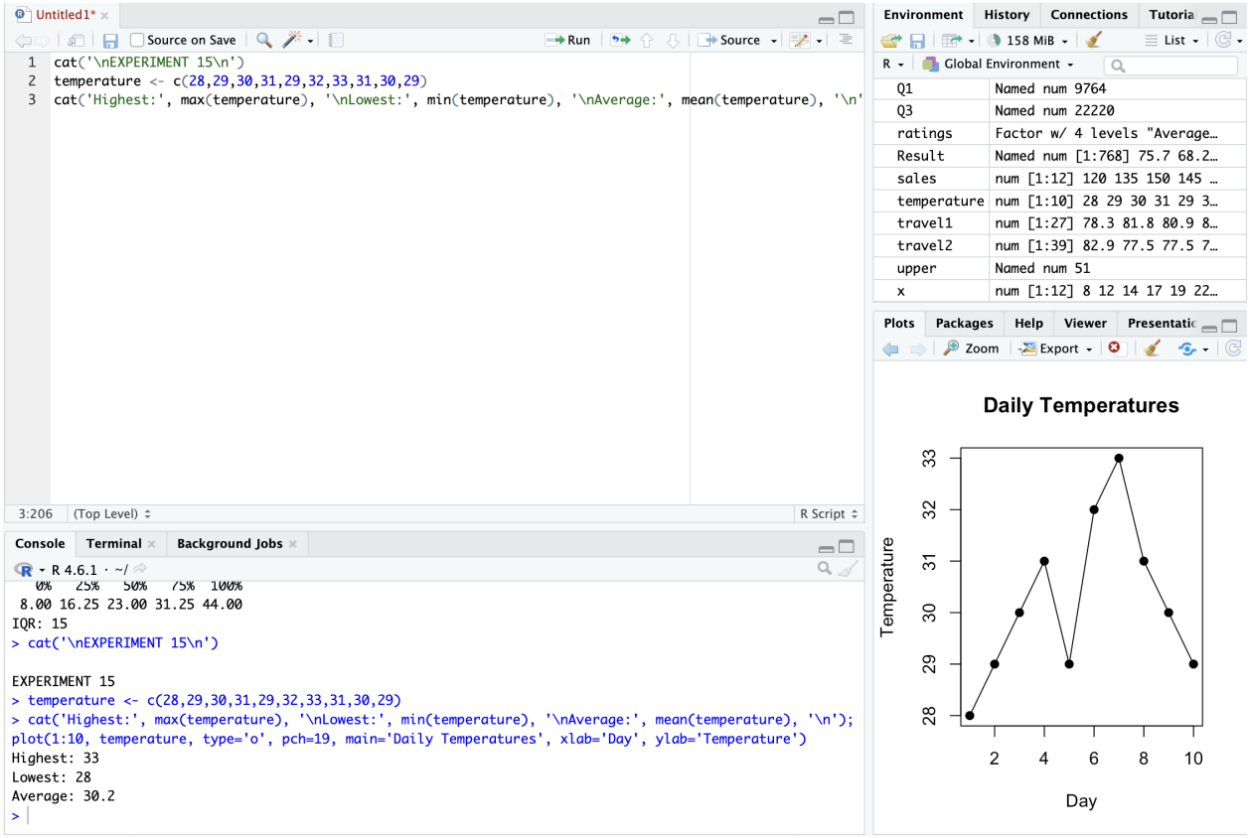
The quartiles and the interquartile range were successfully calculated using R.

## EXPERIMENT 15: Temperature Analysis Using Summary Values and Line Plot

### AIM:

To find the highest, lowest and average temperature and draw a line plot of the daily temperatures using R.

### OUTPUT:




### RESULT:

The highest, lowest and average temperatures were successfully calculated and the line plot was successfully drawn using R.